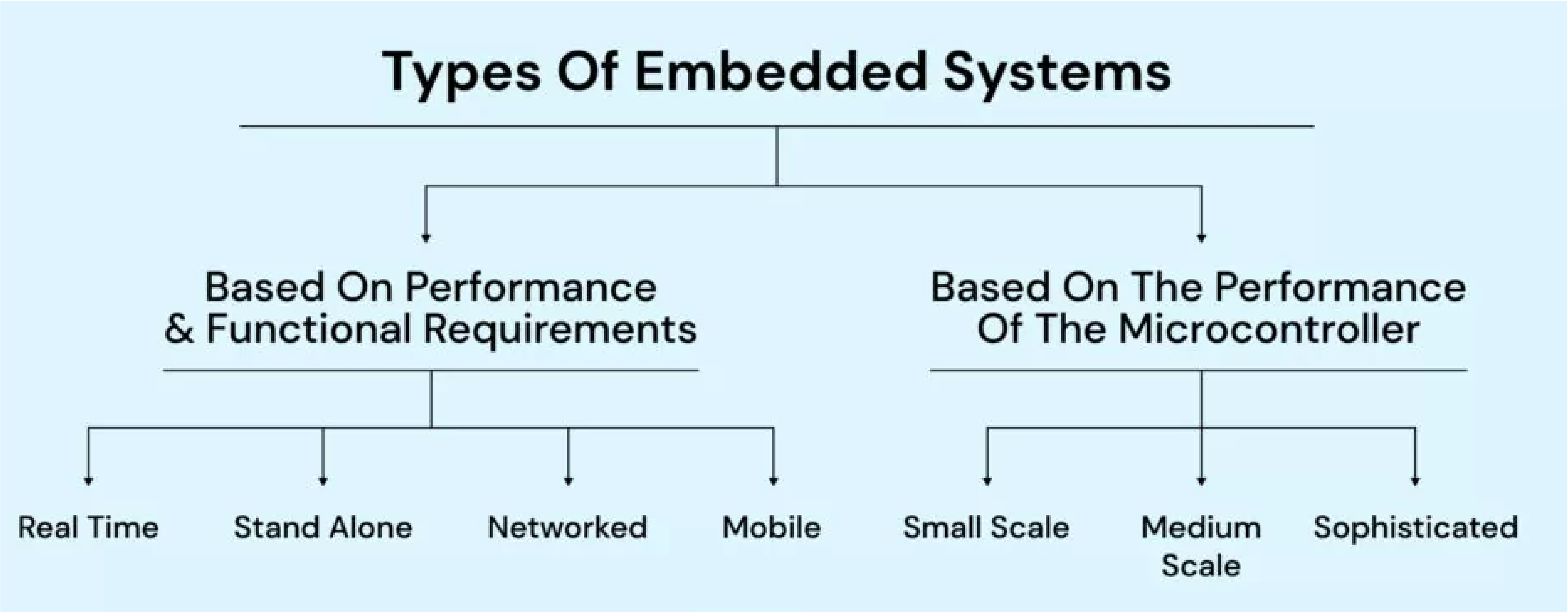
Classification of a Embedded System

Embedded system are classified into two types i.e based on performance & Functional requirements and based on performance of the controller as shown below:



Categories of Embedded Systems

Based on functionality and performance requirements, embedded systems can be categorized as:

1. **Stand-alone embedded systems**
2. **Real-time systems**
3. **Networked information appliances**
4. **Mobile devices**

Classification of a Embedded System

➢ Real-time embedded systems

- This is a complicated hybrid type of embedded solution that brings a range of devices and technologies together under a one combined system.
- It requires stable, high-capacity communication channels to collect environmental data in real time through a network of sensors and deliver it to a centralized node, often supported by AI, to manage system response.
- **In the context of healthcare**, real-time embedded systems usually consist of IoT-connected devices, wearables, and medical equipment deployed in hospital facilities.

Classification of a Embedded System

- **Real-time embedded systems:**
- **Embedded systems in which some specific work has to be done in a specific time period are called real-time systems.**
- **For example, consider a system that has to open a valve within 30 milliseconds when the humidity crosses a particular threshold. If the valve is not opened within 30 milliseconds, a catastrophe may occur. Such systems with strict deadlines are called hard real-time systems.**
- **In some embedded systems, deadlines are imposed, but not adhering to them once in a while may not lead to a catastrophe. For example, consider a DVD player. Suppose, you give a command to the DVD player from a remote control, and there is a delay of a few milliseconds in executing that command. this delay won't lead to a serious implication. Such systems are called soft real-time systems. But, this delaywont lead to a seroiuos implication,. Such systems are called soft real-time systems.**

Classification of a Embedded System

- In this kind of system
  - a great number of sensors, actuators, and other single-purpose devices are interconnected and synchronized.
  - a variety of interfaces and technologies are used to serve one or several purposes.
  - AI or complex algorithms can be used for emergency decision-making.
- **For example,** complicated systems of this type can be used for patient health monitoring and robotic/automated medical care without immediate medical staff interventions. Certain actions, like automated injection of medication, can be performed automatically once sufficient data is collected and verified.

Classification of a Embedded System

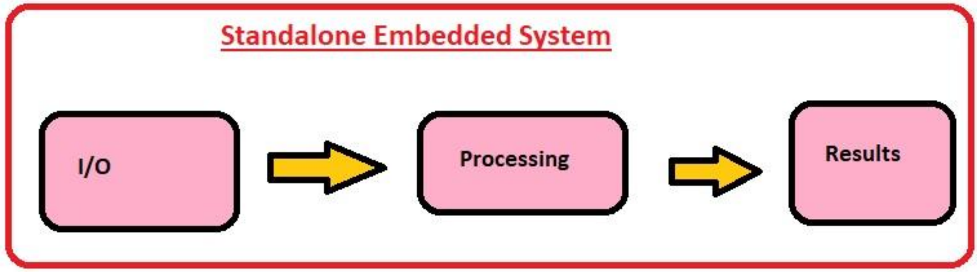
- **Standalone embedded systems**

- This system type doesn’t require a computing device to function.
- As the name suggests, they perform independently.
- In the context of **healthcare**, examples of biomedical standalone embedded systems include:
  - Electronic thermometers
  - Electronic pulse-oximeters
  - Fitness trackers
  - Glucometers
  - Tonometers

Classification of a Embedded System

- **Standalone embedded systems**

Standalone embedded systems are **self-contained, independent computing devices designed for specific tasks, working without a host computer, taking inputs (digital/analog) to provide outputs, and found in everyday items like digital watches, calculators, cameras, and appliances, focusing on efficiency for their dedicated function**



Classification of a Embedded System

Key Characteristics

•**Independence**: They operate autonomously, not relying on a separate computer.
•**Dedicated Function**: Optimized to perform one or a few specific tasks reliably and efficiently.
•**Integrated**: Combine hardware (processor, memory, peripherals) and software for their purpose.
•**Input/Output**: Process inputs (sensors, buttons) to produce outputs (displays, actuators).

Examples

•**Consumer Electronics**: Digital cameras, MP3 players, calculators, smartwatches.
•**Home Appliances**: Microwaves, refrigerators, washing machines (the control panel/logic).
•**Measurement**: Temperature monitoring systems

How They Work (Simplified)
Input: Receive data from sensors or user interfaces (e.g., temperature sensor, keypad).

Processing: A microcontroller or microprocessor executes embedded software to perform calculations or logic.

Output: Generate a result, controlling a display (LCD), light (LED), or motor.

Classification of a Embedded System?

- **Networked embedded systems**

- Networked embedded systems perform specialized tasks by communicating across networks,
- Networked Embedded Systems are connected to a network which may be wired or wireless to provide output to the attached device. They communicate with embedded web server through network. **Examples :**Home security systems, ATM machine, Card swipe machine

Classification of a Embedded System?

- **Networked embedded systems**

- This type of biomedical embedded system relies on local networks and web communications for operation and data transmissions.
- Usually, the following examples of embedded systems belong to the “networked” category:
  - Assisted living or memory care surveillance systems and fall-detection tools
  - Automated surgery and medical service robots
  - Laboratory automation solutions and test machines
  - IoT-based inpatient health monitoring systems with sensors, cameras, alarms, etc.

Classification of a Embedded System

- Mobile embedded systems are small and easy to use and requires less resources. They are the most preferred embedded systems. In portability point of view mobile embedded systems are also best.
- Examples :
- MP3 player
- Mobile phones
- Digital Camera
- le embedded systems

Classification of a Embedded System

- **Mobile embedded systems**

- Portable embedded solutions include handheld devices (for example, a tablet or smartphone) and some kind of biosensors synchronized with it.
- Examples :
  - An ECG-monitoring biosensor connected to the server through web/mobile communication channels.
  - A tablet application for physicians responsible for monitoring ECG events and reviewing cardiograms for each patient.

Classification of a Embedded System

- **Based on Performan ce and micro-controller it is divided into 3 types as follows**
- **1.Small Scale Embedded Systems :**
- 2Medium Scale Embedded Systems :
- 3Sophisticated or Complex Embedded Systems

Classification of a Embedded System

- **Small Scale Embedded System**

- It is designed with single 8 bit or 16bit Microcontroller with little hardware and software complexity.
- They May even be battery operated.
- Usually “C” is used for developing these system.
- The need to limit power dissipation when system is running continuously.
- **Programming tools: Editor, Assembler and Cross Assembler**

Classification of a Embedded System

➢ **Medium Scale Embedded System**

- It is designed with single or few 16 or 32 bit microcontrollers or Digital Signal Processors (DSP) or Reduced Instructions Set Computers (RISC).
- **Programming tools: RTOS, Source code Engineering Tool, Simulator, Debugger and Integrated Development Environment (IDE).**
- Medium Scale Embedded Systems are designed using an 16-bit or 32-bit micro-controller. These medium Scale Embedded Systems are faster than that of small Scale Embedded Systems. Integration of hardware and software is complex in these systems. Java, C, C++ are the programming languages are used to develop medium scale embedded systems. Different type of software tools like compiler, debugger, simulator etc are used to develop these type of systems.

Classification of a Embedded System

- **Sophisticated Embedded System**

- Enormous hardware and software complexity which may need scalable processor or configurable processor and programming logic arrays.
- Constrained by the processing speed available in their hardware units.

- **Programming Tools: For these systems may not be readily available at a reasonable cost or may not be available at all. A compiler or retargetable compiler might have to be developed for this.**

- Sophisticated or Complex Embedded Systems are designed using multiple 32-bit or 64-bit micro-controller. These systems are developed to perform large scale complex functions. These systems have high hardware and software complexities. We use both hardware and software components to design final systems or hardware products.